
EEE5776. Applied Machine Learning



Module 2. Supervised Learning and Experimental Design

Midterm exam review

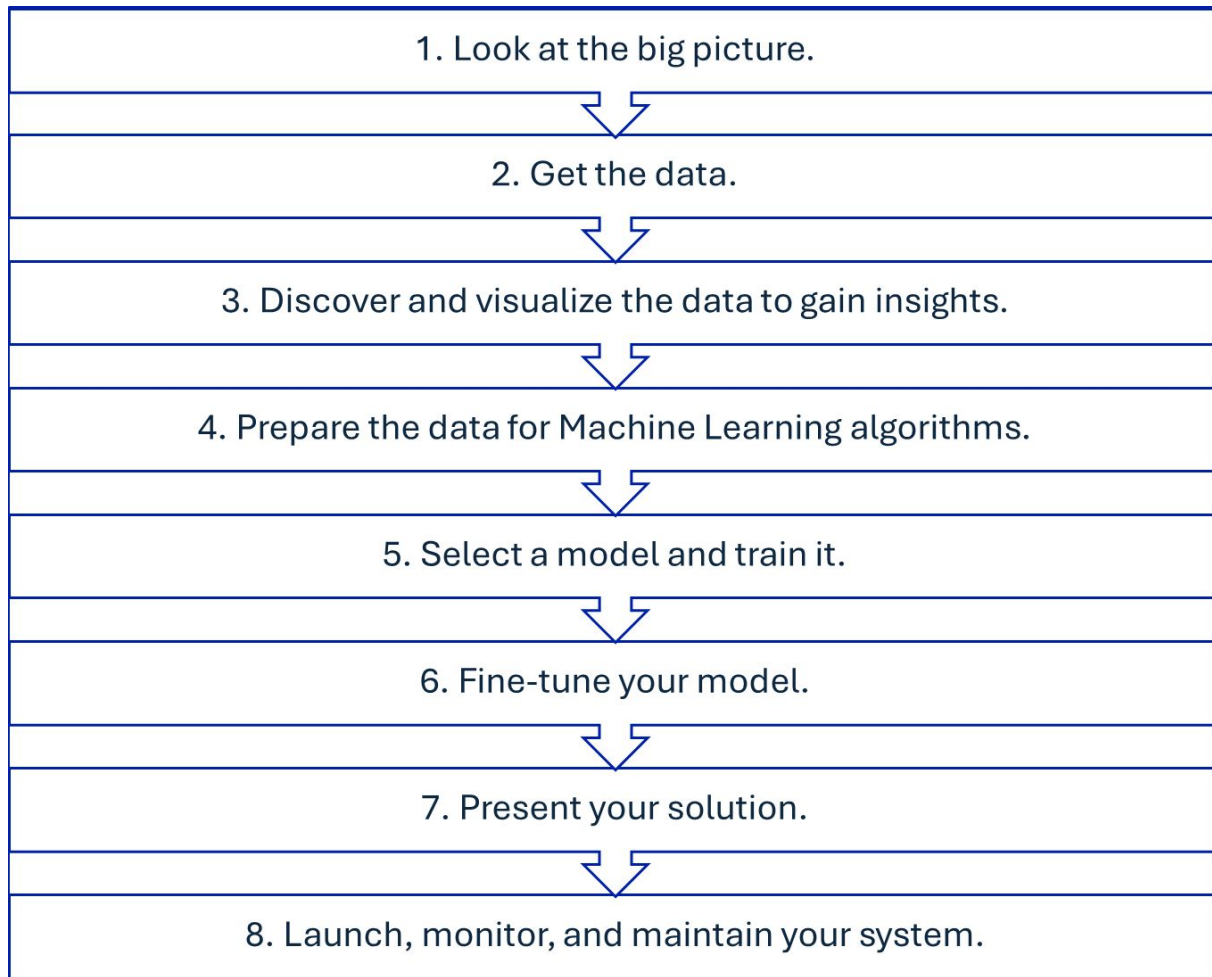
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Midterm exam review

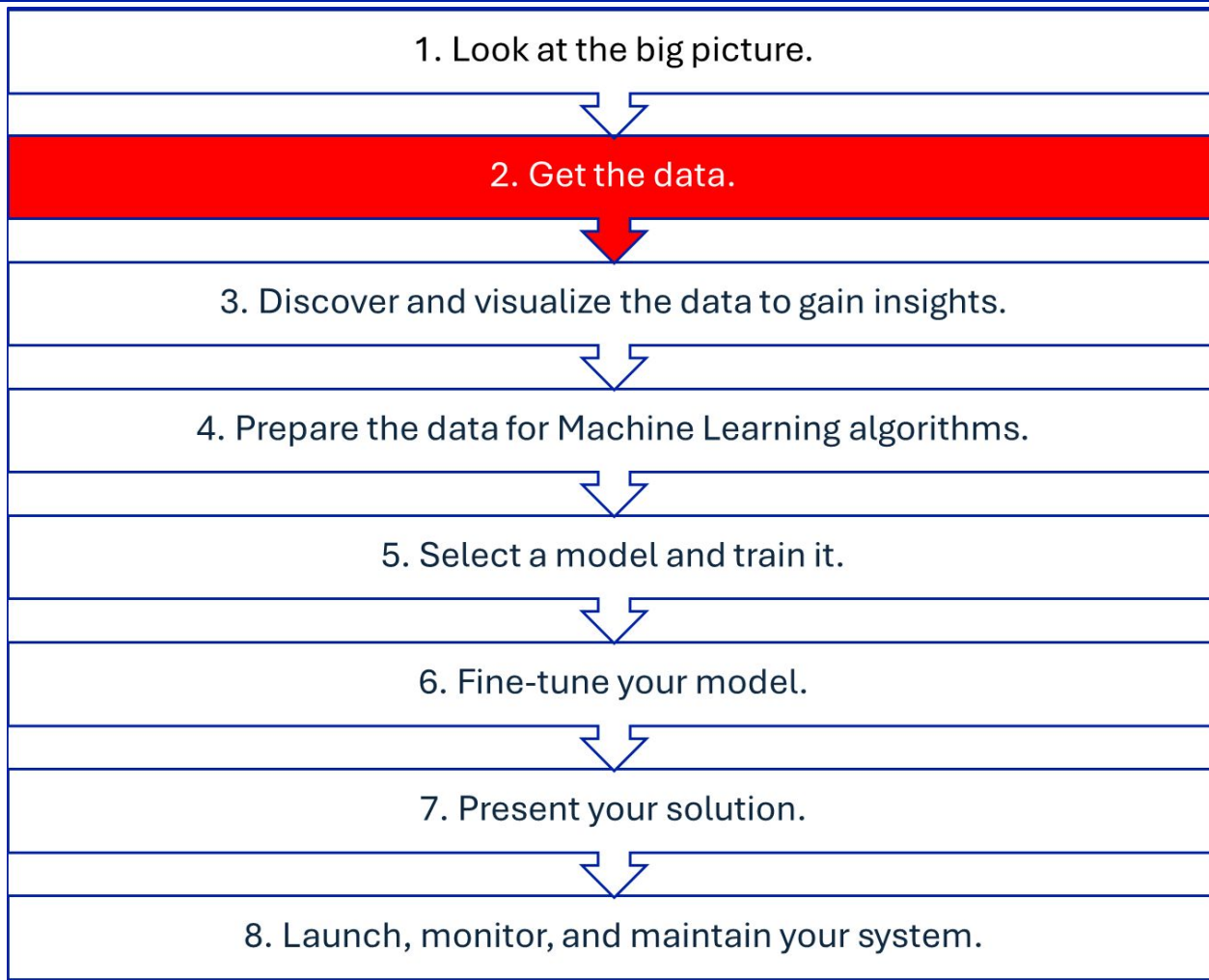
Topics covered

1. AI big picture
2. Machine Learning Pipeline Steps
3. Supervised Learning Tasks
 - a. Regression - Multiple Linear Regression and Polynomial Features
 - b. Classification - Logistic Regression and Decision Trees

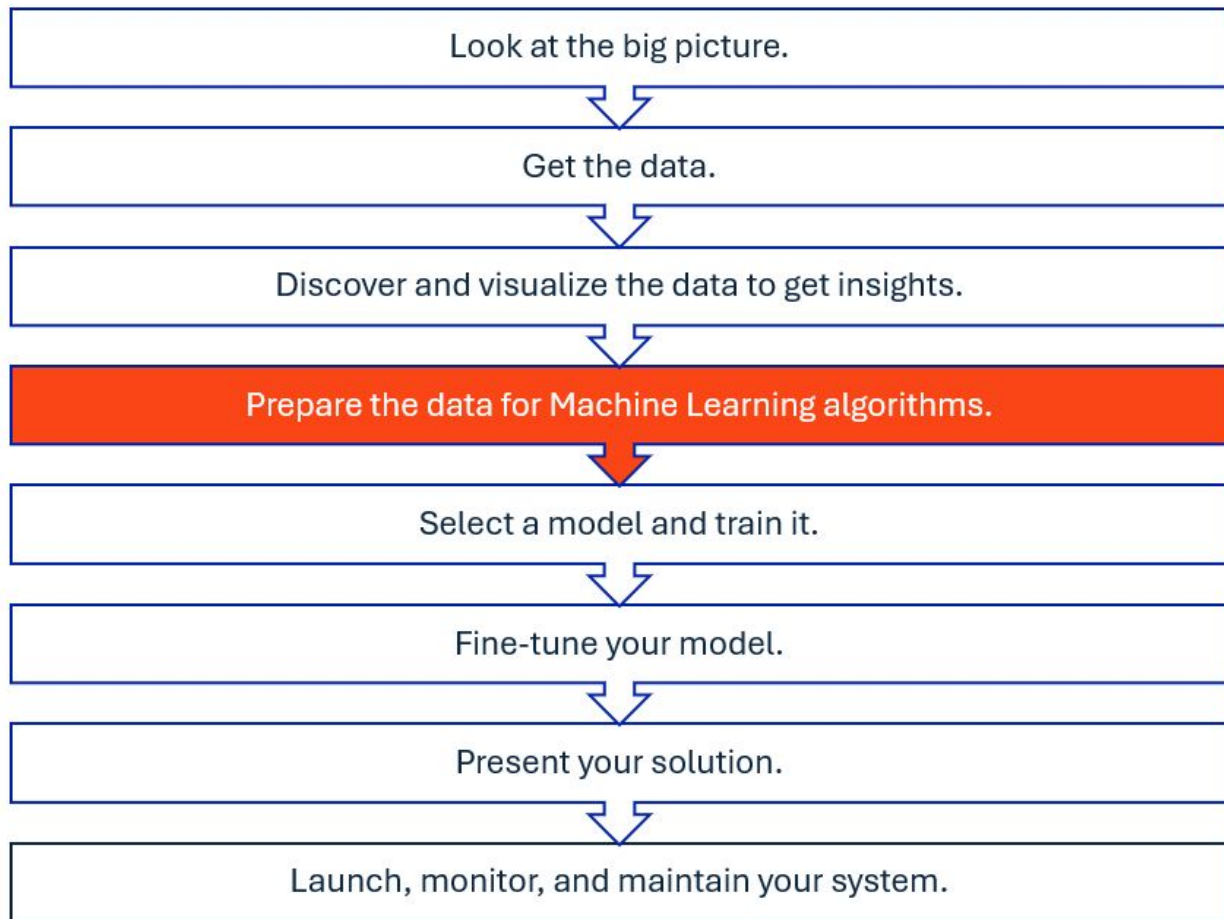
End-to-End Machine Learning Process



End-to-End Machine Learning Process



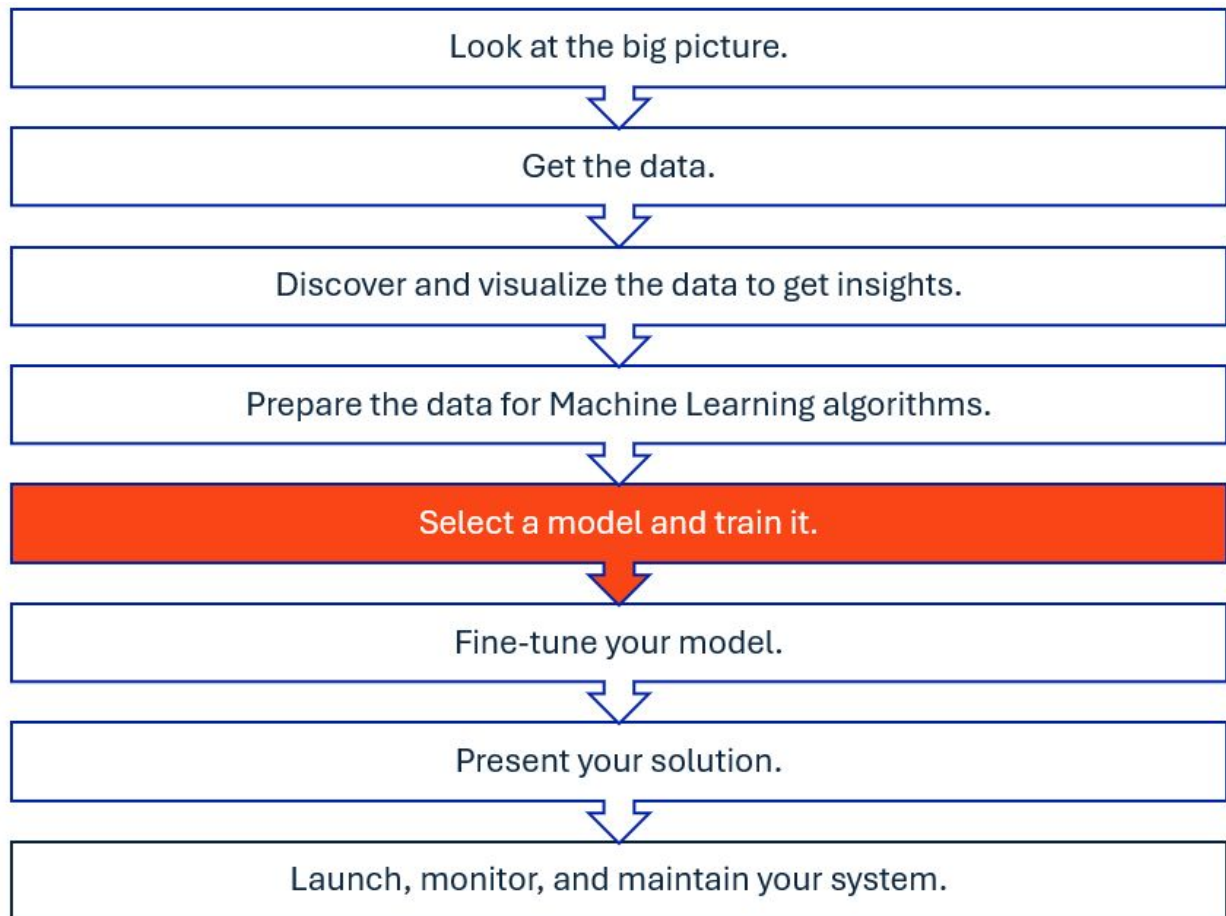
End-to-End Machine Learning Process



Preparing the Data

- Partitioning the Data
- Data cleaning. Missing values and Outliers
- Feature Scaling and encoding categorical values
- Transformation Pipelines
 - Stratification

End-to-End Machine Learning Process



Challenges in Training

- Overfitting vs. Underfitting
 - Cross-validation
 - Regularization

Let's go deeper - Linear regression

1. **Mapper:** a linear combination of features weighted by a set of parameters

$$y = f(\phi(x), \mathbf{w}) = \sum_{j=0}^M w_j \phi_j(x)$$

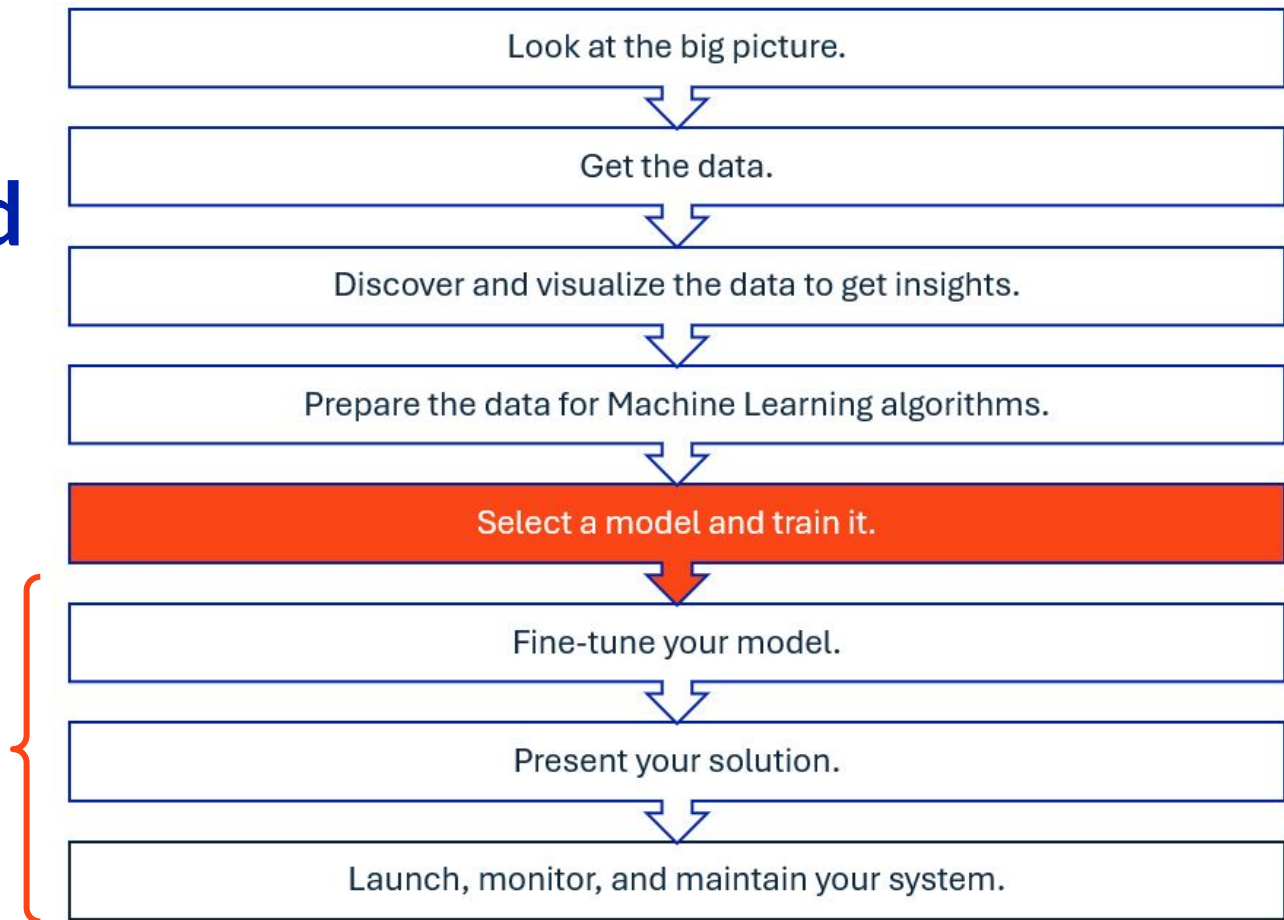
2. **Objective Function:** function that assesses the quality of the output of the model, for example, an error-based function such as L2-norm of the error or the mean-squared error (MSE) function:

$$J(w) = L(w) + \lambda R(w)$$

3. **Learning Algorithm:** an algorithm that solves or searches for the optimal values of the parameters $\mathbf{w}$ that optimize the objective function. It solves the following optimization problem:

$$\arg_{\mathbf{w}} \min J(\mathbf{w})$$

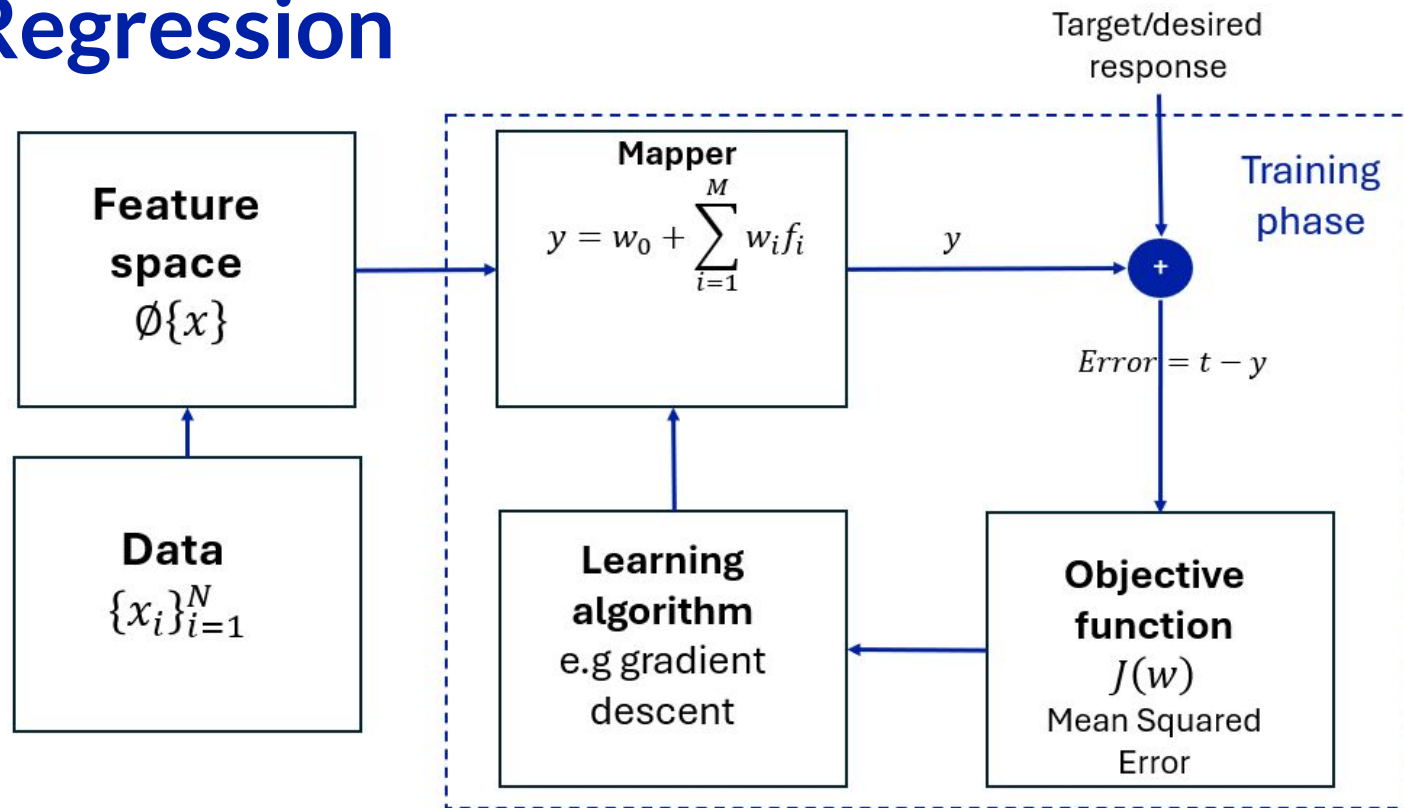
End-to-End Machine Learning Process



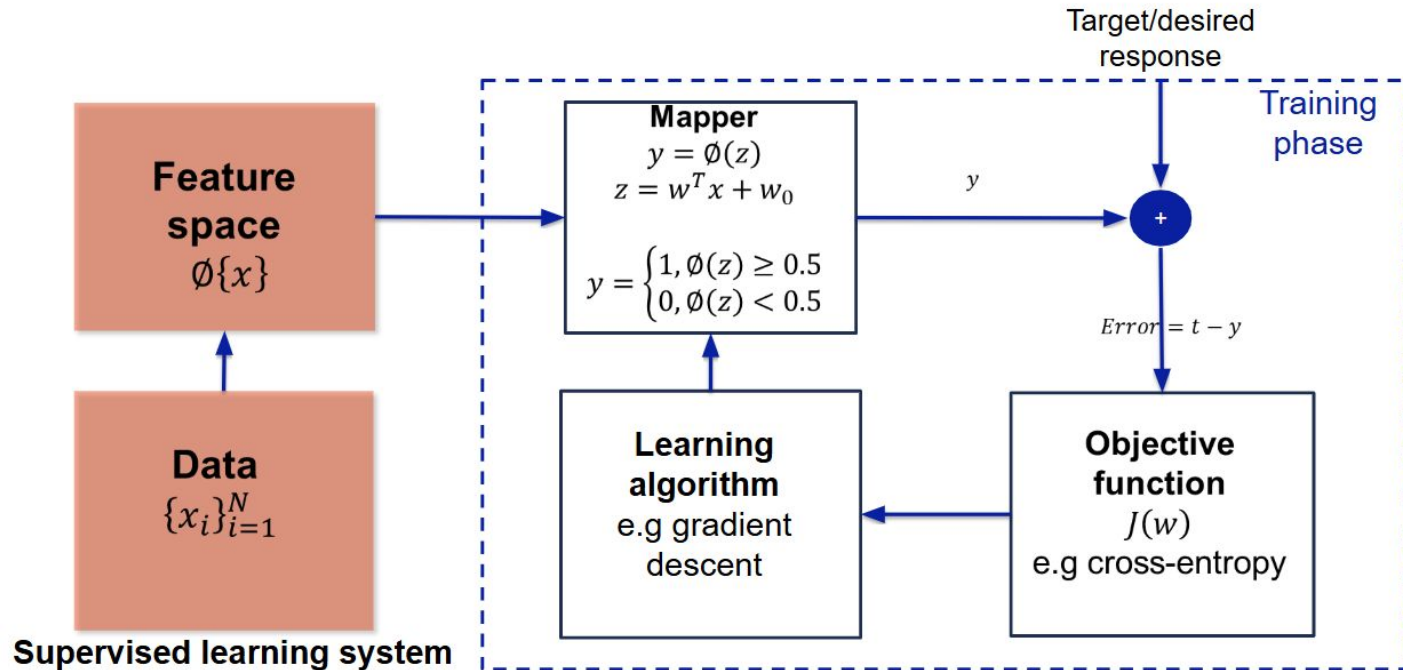
Final steps

- Hyperparameter Tuning
- Evaluate in Test Set
- Present Solution

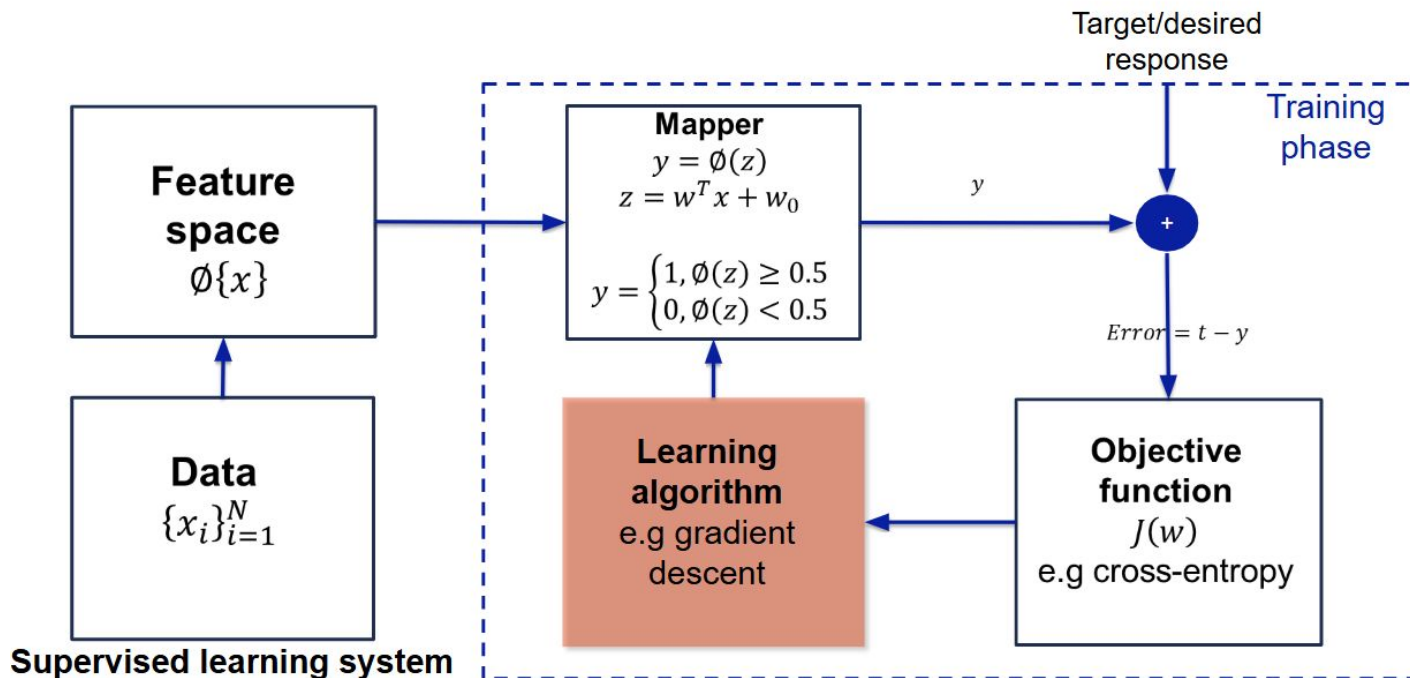
Supervised Learning - Multiple Linear Regression



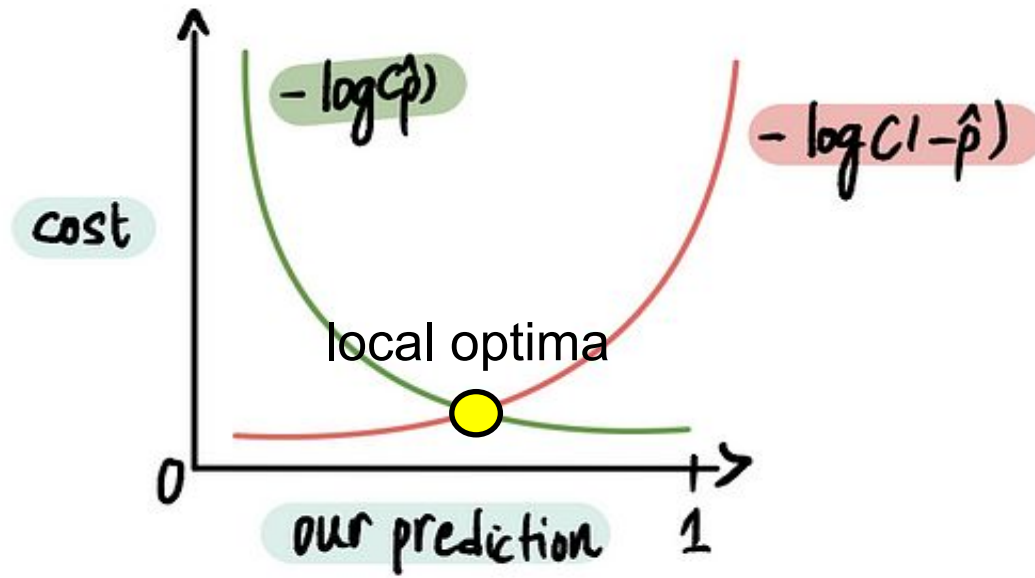
Supervised Learning Process - Logistic Regression Example



Supervised Learning Process - Logistic Regression Example



Supervised Learning Process - Logistic Regression Example



Metrics for Performance

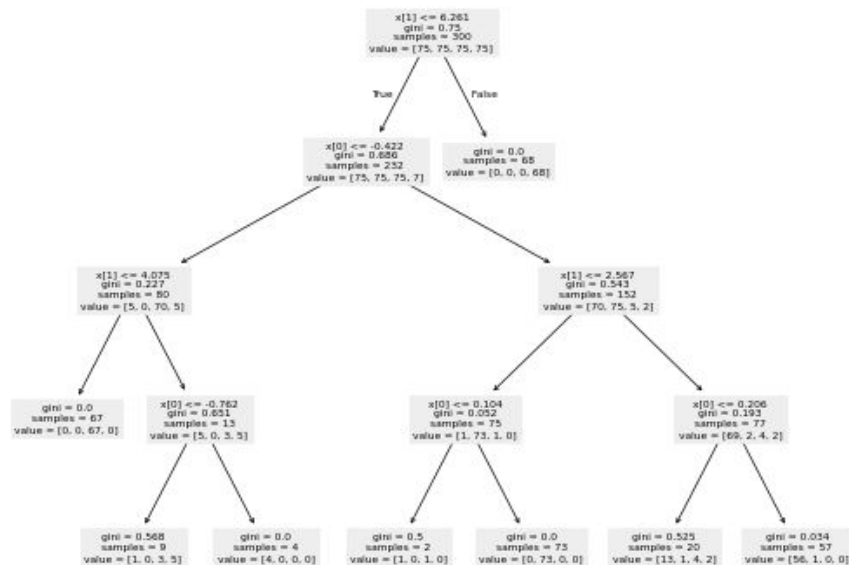
- Regression
 - MSE
 - RMSE
 - R^2
 - ...
- Classification
 - Qual metrics: confusion matrix, precision vs recall plot, AOC
 - Quan metrics: precision, recall, f1., etc

Decision Tree

- **Non-parametric discriminative classifier**
- The process of fitting a decision tree to our data can be done in Scikit-Learn with the `DecisionTreeClassifier` estimator

```
dt = DecisionTreeClassifier(max_depth=4, random_state=0)
dt.fit(X,t)

tree.plot_tree(dt);
```



Impurity and Information Criteria

Each split uses the feature that maximizes the information gain:

The weighted
sum of individual
impurity
measures

$$IG(D_p, f) = I(D_p) - \frac{N_{\text{left}}}{N_p} I(D_{\text{left}}) - \frac{N_{\text{right}}}{N_p} I(D_{\text{right}})$$

The information gain is going to be the objective function

Project Approaches

- Dataset cleaning and preprocessing approaches
- Classification tasks selected
- Regression tasks selected